

PSU SWMDT 12V 10A

Switch Mode Power Supply Unit DC12V 10A Desktop 120W, DC Plug
5.5*2.1mm. Adaptor Size 168*65*39mm

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+12v in
-12v in
INPUT Voltage

Voltage Reference: AnalogReference
AnalogReference: analogReference(0)

5.0v dc
out

5.0v dc IN

5.0v dc IN

IN

Out

FROM
GPIO4

IN

IN

IN

OUT

OUT

OUT

PIR SENSORS

4Ω DuPont wire 3W

3w Speaker

Complete Pin Connection Table — from the firmware:

GPIO	Pin on ESP32	Connect to	Notes
13	GPIO13	Matrix DATA IN	Main 16×32 display
4	GPIO4	BAR3 strip DATA IN	30-LED status chain (3 strips in series)
21	GPIO21	SDA	DS3231 RTC + BH1750 (shared I ² C bus)
22	GPIO22	SCL	DS3231 RTC + BH1750 (shared I ² C bus)
16	GPIO16	DFPlayer TX → ESP RX1kΩ resistor on DFPlayer RX side	
17	GPIO17	DFPlayer RX ← ESP TX1kΩ resistor in series here	
5	GPIO5	Siren MOSFET gate	10kΩ series + 10kΩ pulldown to GND
19	GPIO19	Button — MODE (red)	
26	GPIO26	Button — SET (blue)	
27	GPIO27	Button — UP (white)	
14	GPIO14	Button — ALARM ON/OFF (white)	
12	GPIO12	Button — CHIMES (red)LOW at boot is safe	
32	GPIO32	Button — VOL UP (blue)	
33	GPIO33	Button — VOL DOWN (yellow)	
34	GPIO34	Button — SOUND (yellow)	External 10kΩ pull-up to 3.3V required (input-only pin)
35	GPIO35	Button — ALARM SELECT (red)	External 10kΩ pull-up to 3.3V required (input-only pin)
2	GPIO2	Button — WIFI RESET (blue)Hold 3s: WiFi/network reset. Short tap: Smoker quick-peek (when running)	
18	GPIO18	Button — MEDICINE (yellow)	
25	GPIO25	Button — BIRTHDAY (white)	
15	GPIO15	PIR Courtyard BAM312 direct, 3.3V power	
36	GPIO36	PIR Courtyard CInput-only, no internal pull	
39	GPIO39	PIR Front Door (Zone 2)Input-only, no internal pull	
23	GPIO23	PIR Passage (Zone 2)	
0	GPIO0	—DO NOT USE— boot strap pin	
1	GPIO1	—DO NOT USE— USB serial TX	
3	GPIO3	—DO NOT USE— USB serial RX / Ch340	

Power rails summary:

Rail	Voltage	Feeds
Buck converter out	5.0V, 8A	ESP32 board, DFPlayer, all LEDs
12V PSU	12V	Siren speakers (via IRLZ44N MOSFETs on GPIO5)
Battery backup	12V	Keeps 12V rail alive during loadshedding
ESP32 3.3V out	3.3V	DS3231, BH1750, AM312 PIR sensors, button pull-ups

Connect Dupont jumper wires according
to the wiring table